

Relativity: Essential Computational Methods

A practical reference: each method states what it is for, then the steps to carry it out. Conventions: signature (+ - - -), $\mu \equiv GM/c^2$, an overdot is $d/d\tau$ (or d/du along a curve).

1. Applying a Lorentz transformation / boost

When to use: Relating coordinates, velocities, or any 4-vector between two inertial frames in relative motion.

1. Identify the boost velocity $\vec{v} = \beta c$ between the frames and compute $\gamma = (1 - \beta^2)^{-1/2}$.
2. If the motion is along an axis, use the standard form $ct' = \gamma(ct - \beta x)$, $x' = \gamma(x - \beta ct)$, transverse components unchanged.
3. For a boost in a general direction, split the vector into parts $\parallel$ and $\perp$ to $\vec{\beta}$: the parallel part transforms, the perpendicular part does not. The matrix is

$$\Lambda = \begin{pmatrix} \gamma & -\gamma\beta_j \\ -\gamma\beta_i & \delta_{ij} + \frac{\gamma-1}{|\vec{\beta}|^2}\beta_i\beta_j \end{pmatrix}.$$

4. Apply the *same* Λ to any 4-vector $(x^\mu, u^\mu, p^\mu, a^\mu, \dots)$. To go back, send $\vec{v} \rightarrow -\vec{v}$.
5. *Check:* the interval $c^2t^2 - |\vec{x}|^2$ (and any $A^\mu A_\mu$) must be unchanged.

2. Using invariants instead of transforming

When to use: Collision and decay problems, threshold energies, Doppler shifts — anytime brute-force component transforms would be messy.

1. Build the relevant 4-vectors (typically 4-momenta $p^\mu = (E/c, \vec{p})$).
2. Form a Lorentz scalar by contracting: $p \cdot p = m^2c^2$, or $(p_1 + p_2) \cdot (p_1 + p_2)$, etc. Its value is the same in *every* frame.
3. Evaluate the scalar in whichever frame makes it trivial — usually the centre-of-momentum (CM) frame for thresholds, or the rest frame of one particle.
4. Equate the two evaluations to solve for the unknown.
5. *Key facts:* $E^2 - |\vec{p}|^2c^2 = m^2c^4$; a photon has $p \cdot p = 0$; a single free particle cannot emit/absorb one photon (the invariant forbids it).

3. Computing proper time / time dilation

When to use: Finding the elapsed time on a clock following a given worldline (twin-paradox-type problems, decaying particles, orbiting clocks).

1. Write the metric and the worldline $x^\mu(\lambda)$.
2. Form $d\tau^2 = \frac{1}{c^2} g_{\mu\nu} dx^\mu dx^\nu$ along the path.
3. In flat space this reduces to $d\tau = dt/\gamma = (1 - v^2/c^2)^{1/2} dt$.
4. Integrate $\Delta\tau = \int d\tau$ over the segment of interest.

5. Compare two worldlines between the same pair of events: the one with greater proper time is the “straighter” (geodesic) path.

4. Computing Christoffel symbols

When to use: The first step before geodesics, covariant derivatives, or curvature in any curved space or curvilinear coordinates.

1. *Direct formula:* $\Gamma_{bc}^a = \frac{1}{2}g^{ad}(\partial_b g_{dc} + \partial_c g_{db} - \partial_d g_{bc})$.
2. *Diagonal-metric shortcut* (no summation): off-diagonal $\Gamma_{bc}^a = 0$ for distinct a, b, c ; $\Gamma_{aa}^b = -\frac{1}{2g_{bb}}\partial_b g_{aa}$; $\Gamma_{ba}^a = \partial_b \ln \sqrt{|g_{aa}|}$.
3. *Lagrangian shortcut (often fastest):* write $L = g_{ab}\dot{x}^a\dot{x}^b$, form the Euler–Lagrange equations, rearrange into the form $\ddot{x}^a + \Gamma_{bc}^a\dot{x}^b\dot{x}^c = 0$, and read off the Γ ’s by inspection.
4. Exploit symmetry $\Gamma_{bc}^a = \Gamma_{cb}^a$ to halve the work.

5. Finding geodesics

When to use: Trajectories of free particles (timelike) and light rays (null); orbits around masses.

1. Write $L = g_{ab}\dot{x}^a\dot{x}^b$ using a convenient parameter (proper time for massive, affine parameter for null).
2. Apply Euler–Lagrange $\frac{d}{d\tau}\left(\frac{\partial L}{\partial \dot{x}^a}\right) = \frac{\partial L}{\partial x^a}$ for each coordinate.
3. For any coordinate the metric does *not* depend on, the corresponding momentum $g_{ab}\dot{x}^b$ is conserved — use these first integrals instead of the full second-order equation.
4. Use the normalisation $g_{ab}\dot{x}^a\dot{x}^b = c^2$ (massive) or $= 0$ (null) as an additional first integral.
5. Combine the conserved quantities and normalisation to reduce to a single equation (e.g. an energy equation in r).

6. Solving orbits via the effective potential

When to use: Schwarzschild orbits: circular orbits, ISCO, perihelion precession, light bending.

1. From the geodesic first integrals identify the conserved energy $k = (1 - 2\mu/r)\dot{t}$ and angular momentum $h = r^2\dot{\phi}$ (equatorial plane).
2. Reduce to the radial energy equation $\frac{1}{2}\dot{r}^2 + V_{\text{eff}}(r) = \frac{1}{2}c^2(k^2 - 1)$, with

$$V_{\text{eff}} = -\frac{GM}{r} + \frac{h^2}{2r^2} \left(1 - \frac{2\mu}{r}\right) \quad (\text{massive}).$$

3. Circular orbits: solve $V'_{\text{eff}}(r) = 0$; stability from the sign of V''_{eff} . (Stable for $r > 6\mu$; ISCO at $r = 6\mu$.)
4. For orbit *shape*, substitute $u = 1/r$ to get $u'' + u - 3\mu u^2 = GM/h^2$ (massive) or $u'' + u = 3\mu u^2$ (null); treat the $3\mu u^2$ term perturbatively for precession ($\Delta\phi = 6\pi GM/[a(1 - e^2)c^2]$) and bending ($\Delta\phi = 4GM/c^2b$).

7. Raising, lowering, and covariant derivatives

When to use: Manipulating tensor expressions correctly and computing derivatives that are themselves tensors.

1. Lower with g_{ab} , raise with g^{ab} ; remember $g^{ab}g_{bc} = \delta_c^a$.
2. For a covariant derivative, start with the partial derivative, then add one $+\Gamma$ term for each upper index and subtract one Γ term for each lower index:

$$\nabla_a v^b = \partial_a v^b + \Gamma_{ac}^b v^c, \quad \nabla_a X_b = \partial_a X_b - \Gamma_{ab}^c X_c.$$

3. Use shortcuts where possible: divergence $\nabla_a v^a = \frac{1}{\sqrt{|g|}} \partial_a (\sqrt{|g|} v^a)$; Laplacian $\nabla^2 \phi = \frac{1}{\sqrt{|g|}} \partial_a (\sqrt{|g|} g^{ab} \partial_b \phi)$.
4. *Check*: contracting a free upper with a free lower index should give a scalar; the metric passes through any covariant derivative ($\nabla_a g_{bc} = 0$).

8. Computing curvature & deciding if a space is flat

When to use: Testing for genuine curvature (vs. a bad coordinate choice) and feeding the Einstein equations.

1. Compute the Γ 's (Method 4), then the Riemann tensor

$$R_{abc}{}^d = -\partial_a \Gamma_{bc}^d + \partial_b \Gamma_{ac}^d + \Gamma_{ac}^e \Gamma_{be}^d - \Gamma_{bc}^e \Gamma_{ae}^d.$$

2. Use symmetries to limit work: only $N^2(N^2 - 1)/12$ independent components (1 in 2D, 6 in 3D, 20 in 4D).
3. *Flatness test*: the space is flat iff every component of $R_{abc}{}^d$ vanishes — a singular-looking metric may still be flat (e.g. Minkowski in odd coordinates).
4. Contract for the Ricci tensor $R_{ab} = R_{cab}{}^c$ and scalar $R = g^{ab} R_{ab}$; form $G_{ab} = R_{ab} - \frac{1}{2} g_{ab} R$.

9. Identifying symmetries with Killing vectors

When to use: Finding conserved quantities along geodesics without solving the equations of motion.

1. A vector ξ^a is Killing if $\nabla_a \xi_b + \nabla_b \xi_a = 0$.
2. Shortcut: if the metric is independent of a coordinate x^c , then $\partial/\partial x^c$ is a Killing vector.
3. For any geodesic with tangent t^a , the quantity $\xi_a t^a$ is conserved.
4. Use these conserved quantities directly as the first integrals in Methods 5–6.

10. Setting up the Einstein equations & weak-field limit

When to use: Connecting geometry to matter, and recovering Newtonian gravity as a check.

1. Build the stress-energy tensor; for a perfect fluid $T^{\mu\nu} = (\rho + p/c^2) u^\mu u^\nu - p g^{\mu\nu}$.
2. Write the field equations $G_{\mu\nu} = -\kappa T_{\mu\nu}$ with $\kappa = 8\pi G/c^4$ (add $\Lambda g_{\mu\nu}$ on the left if needed); in vacuum simply $R_{\mu\nu} = 0$.
3. *Weak field*: put $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ with $|h| \ll 1$ and identify $g_{00} \approx 1 + 2\Phi/c^2$.
4. Keep leading order to recover Poisson's equation $\nabla^2 \Phi = 4\pi G \rho$ — this fixes the constant κ and confirms the Newtonian limit.

These ten methods cover the bulk of the example-sheet calculations. The recurring strategy: exploit symmetry and invariants to avoid brute-force work, then use a normalisation or limit as an independent check.